

SANTA MARIA, CA, USA

WMO: 723940

Lat: 34.899N

Lon: 120.449W

Elev: 74

StdP: 100.44

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23273

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.9	2.1	-6.0	2.3	13.1	-3.9	2.8	10.4	10.6	14.8	9.2	14.5	1.3	120	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	11.3	29.5	16.8	27.2	16.6	25.2	16.3	19.2	25.8	18.3	24.6	17.6	23.5	4.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.1	21.0	15.8	11.3	20.3	15.2	10.9	19.6	54.6	25.9	51.9	24.4	49.8	23.4	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.2	8.2	DB	-1.7	35.0	1.4	2.5	-2.7	36.8	-3.5	38.3	-4.3	39.7	-5.3	41.5	(4)
(5)			WB	-2.9	20.8	1.4	1.4	-3.9	21.7	-4.7	22.5	-5.5	23.3	-6.5	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	11.6	11.9	13.0	13.5	14.8	16.1	17.5	17.7	17.9	16.6	13.9	11.3	(6)	
	DBStd	3.45	2.77	2.95	2.70	2.72	2.17	1.89	1.75	1.71	2.37	2.91	3.01	2.70	(7)	
	HDD10.0	57	15	12	5	3	0	0	0	0	0	0	4	17	(8)	
	HDD18.3	1419	208	182	167	147	113	71	37	32	36	71	138	218	(9)	
	CDD10.0	1761	66	64	97	109	149	183	233	240	236	206	120	58	(10)	
	CDD18.3	82	0	1	1	3	4	4	12	13	22	18	4	0	(11)	
	CDH23.3	944	17	28	37	62	68	61	75	81	195	237	75	9	(12)	
(13)	CDH26.7	268	1	5	7	20	23	18	10	17	63	86	18	0	(13)	
(14)	Wind	WSAvg	3.0	2.6	2.9	3.3	3.6	3.7	3.4	2.9	2.7	2.6	2.7	2.5	2.6	(14)
(15) Precipitation	PrecAvg	319	62	71	59	24	5	1	0	1	4	14	34	47	(15)	
	PrecMax	680	299	294	239	99	62	22	10	22	41	61	120	250	(16)	
	PrecMin	77	0	1	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	137	58	68	52	27	11	3	2	3	7	16	30	42	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.7	27.0	27.2	29.3	29.6	28.5	27.9	28.6	32.4	33.1	29.1	24.5	(19)	
		MCWB	12.8	15.1	15.6	15.9	16.2	18.5	19.1	18.7	18.2	17.3	15.8	12.4	(20)	
	2%	DB	22.4	23.1	23.6	24.5	24.5	24.5	25.3	25.2	28.1	29.0	25.4	21.6	(21)	
		MCWB	12.8	13.8	14.3	14.8	15.9	16.9	18.2	18.2	17.7	15.6	14.1	12.4	(22)	
	5%	DB	19.9	19.9	21.0	21.5	21.6	22.4	23.5	23.6	25.1	25.7	22.5	19.0	(23)	
		MCWB	12.2	13.0	13.6	14.4	15.0	16.2	17.5	17.5	17.5	15.3	13.6	12.0	(24)	
	10%	DB	17.3	17.4	18.6	19.2	19.7	20.9	22.0	22.3	23.0	22.9	20.0	17.0	(25)	
		MCWB	12.1	12.2	13.2	13.5	14.3	15.5	16.7	16.9	16.9	15.1	13.3	11.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	16.7	17.2	17.7	18.3	18.6	20.1	19.9	21.0	19.7	17.2	16.2	(27)	
		MCDB	19.6	22.5	23.9	25.7	25.8	26.5	26.1	26.7	27.9	27.5	23.9	19.5	(28)	
	2%	WB	14.9	15.2	15.7	16.0	16.6	17.4	18.6	18.6	19.1	17.9	16.0	14.9	(29)	
		MCDB	18.3	19.9	21.1	22.3	23.1	23.7	24.7	24.5	25.3	24.4	21.3	17.5	(30)	
	5%	WB	13.9	14.1	14.7	14.9	15.4	16.6	17.7	17.8	18.0	16.9	15.1	13.8	(31)	
		MCDB	17.0	18.3	19.4	20.6	20.8	22.1	23.1	23.3	23.8	22.8	19.8	16.4	(32)	
	10%	WB	12.9	13.0	13.8	13.9	14.6	15.6	16.9	17.0	17.1	15.8	14.2	12.8	(33)	
		MCDB	16.2	16.6	17.8	18.5	19.2	20.4	21.7	21.9	22.3	21.2	18.8	15.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	11.7	11.8	12.0	10.8	10.4	9.9	9.9	11.3	12.8	13.1	12.5	(35)	
		MCDBR	16.4	16.2	16.0	16.2	14.6	12.9	11.8	11.7	15.0	18.2	17.2	15.8	(36)	
	5% WB	MCWBR	9.1	8.9	8.6	8.4	7.4	6.4	5.7	5.5	7.0	8.4	8.5	8.9	(37)	
		MCWBR	11.6	12.7	12.8	13.8	12.5	12.1	11.2	11.1	12.9	14.2	13.3	10.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.320	0.324	0.341	0.352	0.352	0.361	0.347	0.347	0.342	0.304	0.308	(40)	
		taud	2.583	2.543	2.537	2.476	2.465	2.461	2.490	2.497	2.503	2.517	2.599	2.583	(41)	
	Ebn at Noon		911	930	951	944	931	926	915	926	913	891	903	883	(42)	
		Edh at Noon	77	89	97	109	111	112	108	105	100	90	75	73	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.98	3.84	5.26	6.58	7.38	7.71	7.33	6.78	5.82	4.45	3.36	2.66	(44)	
	RadStd		0.24	0.40	0.35	0.41	0.44	0.32	0.24	0.25	0.26	0.30	0.18	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.63	N/A	N/A	+1.03	N/A	N/A	N/A	-214	+236	+35	(46)	
(47)	Regional (27 neighbors)	+0.65	N/A	-1.28	+0.75	N/A	N/A	N/A	-167	+240	+76	(47)	

Nomenclature: See separate page